

Vinayak Suresh Chandavar

Electronics and Communication Engineering
Bachelor of Engineering
Sahyadri College of Engineering and Management, Mangaluru

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GitHub Profile
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TECHNICAL SKILLS AND INTERESTS

Languages: Python, C++, Bash, JavaScript, TypeScript.
AI & Computer Vision: Ultralytics YOLOv8, PyTorch, OpenCV, TensorFlow.js.
Hardware & Edge-AI: NVIDIA Jetson Orin Nano, Intel RealSense Depth Cameras, ESP32.
Web Development & Cloud: React 18, Firebase (Firestore, RTDB), Tailwind CSS, Vite.
Areas of Interest: Edge-AI Deployment, Computer Vision, Autonomous Systems.

EDUCATION

- Sahyadri College of Engineering and Management, Mangaluru** 2027
Bachelor of Engineering in Electronics and Communication Engineering
- Shri Marikamba Government PU College, Sirsi**
Pre-University Course (Class 12) – Dept. of PUE, Karnataka
- Government Marigudi High School, Sirsi**
Secondary School Certificate (Class 10) – KSEEB

PROJECTS

- SmartSeg: Edge-AI Based Plastic Waste Sorting System**   *Emphasis Lab*
Real-time CV system on Jetson Orin Nano to automate MRF conveyor belt sorting at 30+ FPS.
 - Technologies used: YOLOv8, OpenCV, Python, Intel RealSense, Headless Linux.
 - Engineered a zero-domain-gap dataset and a lightweight depth-gating algorithm to bypass GPU memory limits and prevent thermal throttling on edge hardware.
 - Implemented a 7/10 multi-frame supermajority rolling buffer to eliminate single-frame classification hallucinations.
- KRUX: AI-Powered Gamified Smart Bin Ecosystem**   
A full-stack hardware/software ecosystem gamifying recycling to prevent contamination.
 - Technologies used: React 18, Tailwind, Firebase (RTDB, Firestore), TensorFlow.js, ESP32.
 - Built a Progressive Web App (PWA) with on-device ML inference in the browser, ensuring 100% user privacy and zero cloud inference costs.
 - Interfaced the web app with an ESP32 hardware bin using Firebase RTDB to achieve a seamless, zero-latency sensor handshake for fraud detection.
- CareLens: Vision AI Wearable for the Visually Impaired**  
Wearable assistive tech providing 3D semantic understanding and audio navigation cues.
 - Technologies used: PyTorch, YOLOv8, Intel RealSense, pytttsx3, Jetson Orin Nano.
 - Fine-tuned PyTorch models for offline indoor obstacle detection, heavily optimizing dataloader constraints (batch=1) to prevent 'CUDACachingAllocator' OOM crashes.
 - Designed a low-latency offline audio engine using 'pytttsx3' and a Type-C DAC to bypass Bluetooth bottlenecks and deliver conversational audio warnings.

EXPERIENCE & RESEARCH

- Emphasis Lab, SCEM** 2023 - Present
Project Developer / AI Researcher Adyar, Mangaluru
 - Developed and deployed real-time Edge-AI solutions utilizing NVIDIA Jetson hardware and depth-sensing technology.
 - Specialized in resolving low-level Linux memory constraints ('nvlm assert' bugs) to maximize deep learning inference on embedded systems.

ACHIEVEMENTS & HACKATHONS

- Hackathon Participant & Finalist:** Various National & State Level Hackathons.  2023 - 2024